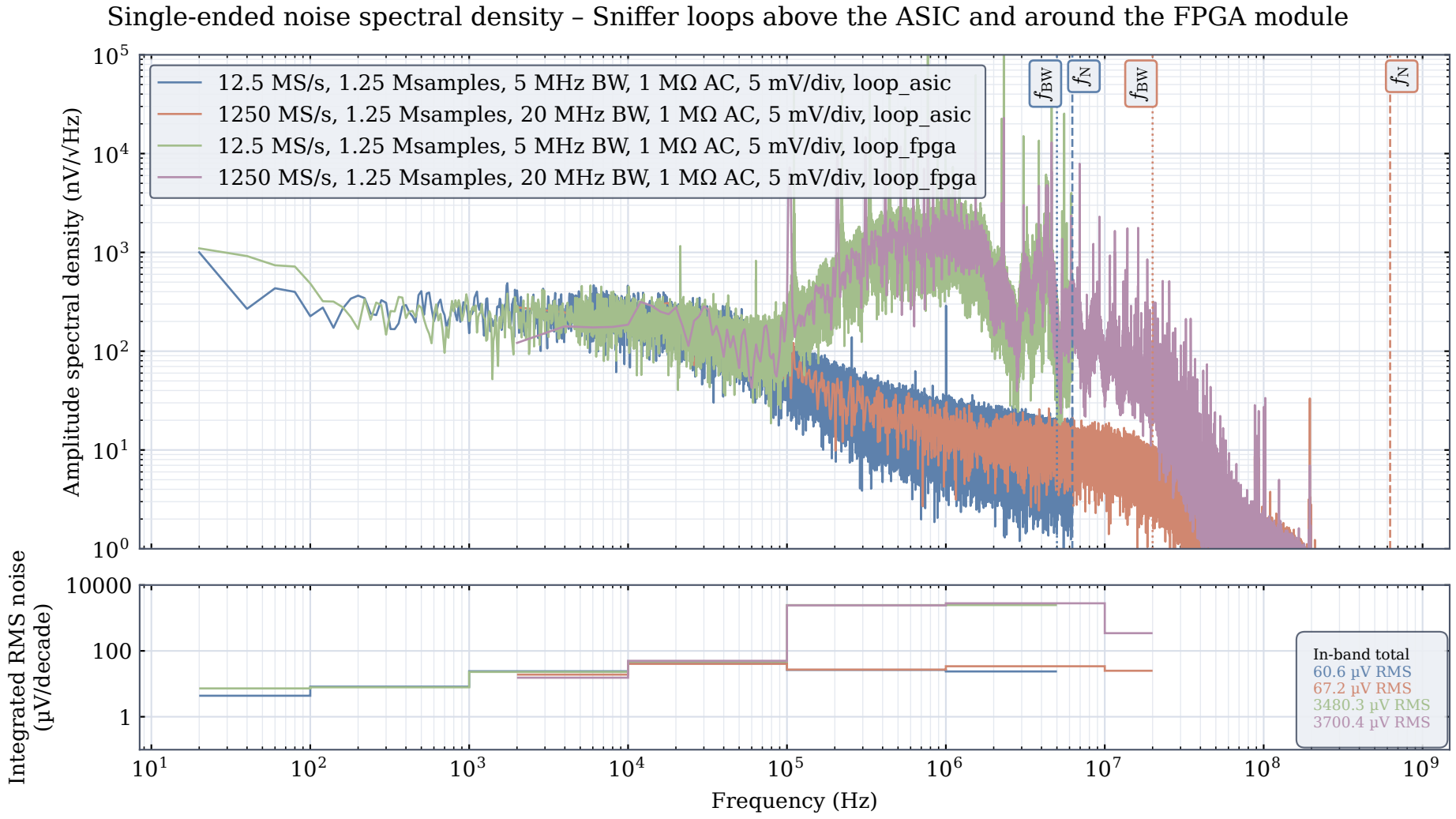

Single-ended noise spectral density – Sniffer loops above the ASIC and around the FPGA module
12.5 MS/s, 1.25 Msamples, 5 MHz BW, 1 MΩ AC, 5 mV/div, loop_asic
1250 MS/s, 1.25 Msamples, 20 MHz BW, 1 MΩ AC, 5 mV/div, loop_asic
12.5 MS/s, 1.25 Msamples, 5 MHz BW, 1 MΩ AC, 5 mV/div, loop_fpga
1250 MS/s, 1.25 Msamples, 20 MHz BW, 1 MΩ AC, 5 mV/div, loop_fpga
f_{BW}
f_N
f_{BW}
f_N
Amplitude spectral density (nV/√Hz)
10^5
10^4
10^3
10^2
10^1
10^0
Integrated RMS noise (µV/decade)
10000
100
1
In-band total
60.6 µV RMS
67.2 µV RMS
3480.3 µV RMS
3700.4 µV RMS
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
Frequency (Hz)